

ANAPA VITYAZEVO, Russian Federation

WMO: 370010

Lat: 45.002N

Lon: 37.347E

Elev: 174

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	15.7	7.1	8.0	17.0	12.1	10.2	19.7	38.0	25.3	34.8	29.0	16.1	50	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.3	92.0	71.7	89.5	71.8	86.3	71.4	77.1	85.8	75.5	84.0	73.9	82.4	12.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	131.2	82.9	73.1	123.5	81.5	71.3	116.2	79.5	40.7	85.6	39.1	84.1	37.5	82.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.8	26.3	22.5	DB	4.9	96.8	9.2	1.9	-1.7	98.2	-7.2	99.3	-12.3	100.4	-19.1	101.8	(4)
(5)			WB	7.9	79.6	7.2	1.7	2.7	80.8	-1.5	81.8	-5.5	82.7	-10.8	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	35.8	37.2	43.1	52.0	61.2	69.5	74.6	75.2	66.2	55.3	46.2	39.8		
	DBStd	15.78	10.76	9.88	7.15	6.33	6.34	5.19	4.69	5.02	6.12	7.91	8.38	9.61		
	HDD50	1633	446	365	233	49	3	0	0	0	0	39	166	332		
	HDD65	4642	906	778	680	393	153	15	0	1	59	313	563	781		
	CDD50	3375	5	7	18	108	349	586	762	783	485	203	53	16		
	CDD65	908	0	0	0	1	34	151	297	319	94	12	0	0		
	CDH74	8553	0	0	1	19	281	1243	2986	3234	715	73	1	0		
	CDH80	2966	0	0	0	1	51	329	1096	1310	167	12	0	0		
	Wind	WSAvg	10.5	12.2	12.8	12.0	11.1	9.1	8.7	9.0	9.5	10.4	10.9	11.7		
	Precipitation	PrecAvg	28.9	3.2	2.5	2.4	2.1	2.6	1.9	2.2	2.1	2.1	2.5	3.1		
		PrecMax	43.0	8.7	5.9	4.4	4.3	3.7	5.3	4.9	6.5	6.7	4.0	5.9		
		PrecMin	21.9	1.1	0.7	1.0	0.2	0.2	0.3	0.3	0.2	0.5	0.4	0.8		
		PrecStd	4.8	1.6	1.2	0.9	1.0	1.1	1.1	1.6	1.3	0.9	1.4	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	61.2	66.4	76.7	84.6	91.8	95.3	96.4	88.2	80.7	70.0	63.0		
		MCWB	48.5	49.8	51.1	60.1	67.8	72.8	72.0	70.4	70.3	66.1	57.9	53.7		
	2%	DB	54.0	56.9	61.1	71.5	80.6	87.4	91.7	93.0	84.3	75.3	65.9	59.0		
		MCWB	47.6	48.5	49.8	57.4	66.2	71.8	72.1	70.8	68.7	63.7	56.7	52.1		
	5%	DB	51.8	53.2	57.2	67.7	77.0	83.8	89.2	89.8	80.9	71.5	62.3	55.7		
		MCWB	46.9	47.2	49.1	56.2	64.6	70.7	72.3	71.6	67.4	62.5	55.9	50.5		
	10%	DB	49.7	49.9	53.7	64.0	73.4	80.7	86.0	87.5	77.9	68.0	59.0	53.3		
		MCWB	46.1	45.5	47.5	53.8	63.5	69.0	71.9	72.0	66.1	60.8	54.3	49.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	51.8	53.7	62.4	71.3	76.6	79.4	79.8	74.8	68.4	60.6	56.3		
		MCDB	54.2	57.0	62.6	72.6	80.3	85.8	87.4	87.5	82.8	76.0	65.9	60.9		
	2%	WB	48.9	49.8	51.2	59.0	68.6	74.5	77.3	77.3	72.2	65.5	58.3	53.0		
		MCDB	52.5	54.8	58.7	68.7	77.1	82.9	85.4	86.2	80.2	72.2	63.3	57.4		
	5%	WB	47.5	47.8	49.8	56.8	66.6	72.6	75.6	75.6	69.9	63.6	56.5	51.3		
		MCDB	51.2	52.0	55.9	65.2	74.3	80.6	83.7	84.4	77.5	70.1	61.4	55.1		
	10%	WB	46.0	45.9	48.3	54.9	64.8	70.7	73.9	73.8	67.9	61.5	54.4	49.6		
		MCDB	49.2	49.5	53.2	62.0	71.8	78.2	82.4	82.5	75.3	67.0	59.2	53.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	13.9	14.9	17.8	18.7	19.1	20.2	21.3	20.0	17.3	15.4	12.9		
		MCDBR	15.1	17.5	20.6	23.7	23.9	22.5	23.4	24.2	23.6	21.9	20.0	15.5		
		MCWBR	11.0	11.1	12.2	13.3	13.3	11.9	11.2	12.0	13.0	12.9	13.5	10.8		
		MCDBR	13.7	15.5	18.4	21.2	20.9	19.6	20.1	20.9	21.0	19.6	18.2	14.6		
	5% WB	MCWBR	10.8	10.8	12.0	12.9	12.5	11.6	11.9	12.7	13.3	12.7	13.4	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.378	0.413	0.447	0.415	0.417	0.438	0.440	0.409	0.385	0.356	0.348		
	taud		2.303	2.232	2.162	2.108	2.303	2.341	2.289	2.267	2.326	2.348	2.360	2.314		
	Ebn at Noon		235	249	256	258	272	271	263	258	256	245	231	221		
	Edh at Noon		26	34	41	47	40	39	41	40	35	30	25	24		
(44) All-Sky Solar Radiation	RadAvg		359	616	1015	1555	1958	2157	2157	1901	1405	872	480	314		
	RadStd		49	82	87	143	161	124	93	124	98	77	55	36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.58	N/A	N/A	+2.57	N/A	N/A	N/A	N/A	+349	+180
(47)	Regional (5 neighbors)	+1.17	N/A	N/A	N/A	N/A	N/A	-214	-301	+261	+157

Nomenclature: See separate page